CYBERSECURITY SKILLS DEVELOPMENT PORTFOLIO

Cyber Talent Summer Camp Challenge

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1. Executive Summary

This portfolio document reflects my learning and developing skills during **the Cyber Talent Summer Camp Challenge.** This skill-based cybersecurity training program, and other programs such as Project based Cyber security, Development, Aerospace, Emerging Technology, and Embedded system are delivered by **the Information Network Security Administration (INSA)** in Five Universities. One of these is Bahirdar University. I attended my training at Bahir Dar University, and the training period was Jun 25 to Aug 24, 2026 G.C.

During the program, I studied **foundational cybersecurity** topics, including **Linux fundamentals and computer networking, and explored security tools such as Nmap, Burp Suite, and Gobuster in learning and laboratory contexts.** I also prepared technical write-ups on topics and exercises including OWASP, Airplane, Blue, networking, and Python code. In addition, There were also Psychology lesson which helps me to identify my skill, and enhance my self-development, as well as problem-solving, and sport session to maintain our health both physically, and mentally.

Alongside the training, I continued independent practice through **TryHackMe and PicoCTF.** I also used **freeCodeCamp** to study Python programming and introductory English-language material at A2 level. These activities supported my technical learning and helped me recognize the value of programming, communication, and continuous practice in cybersecurity.

This portfolio presents a summary of the training, the resources used, selected practical work, and areas for further development. Detailed lab reports and supporting evidence are linked from the accompanying GitHub repository.

2. Introduction

Cybersecurity requires a combination of technical knowledge, practical problem-solving, careful analysis, and clear documentation. My participation in the Cyber Talent Summer Camp Challenge provided an opportunity to develop foundational knowledge and apply it through guided learning activities and hands-on exercises.

The purpose of this document is to record the learning activities I undertook, identify the tools and resources I used, summarize my technical documentation, and outline the skills I intend to strengthen. It is designed to accompany my GitHub portfolio and video presentation.

3. Skills and Knowledge Developed

3.1 Linux Fundamentals

I studied Linux fundamentals and practiced using the command line in learning environments. My continued goal is to become more confident with routine system operations, file management, permissions, processes, and networking-related commands.

3.2 Computer Networking

I studied foundational networking concepts and practiced activities that helped me understand how network services and hosts can be examined in an authorized lab environment. I documented networking-related work to reinforce the concepts and commands used.

3.3 Cybersecurity Tools

Tools explored during my learning included:

- **Nmap** — for learning network discovery, port scanning, and service enumeration in authorized lab environments.
- **Burp Suite** — for learning about web application testing and inspecting HTTP requests and responses in permitted labs.
- **Gobuster** — for learning content and directory discovery in authorized lab environments.
- Operating Systems — like Kali Linux and Ubuntu in virtualized environment. These tools helped me to develop my Linux operating system skills, understand networking concepts, and practice cybersecurity techniques in controlled laboratory environments.
- My objective is to strengthen not only tool familiarity but also my ability to explain the purpose of a command, interpret its output, and select appropriate next steps.

3.4 Technical Documentation and Problem-Solving

Preparing write-ups helped me practice recording objectives, methods, observations, and lessons learned. I intend to improve these reports by adding clear explanations, relevant screenshots, and references to official documentation.

3.5 Python Programming

I studied introductory Python concepts through freeCodeCamp and worked through selected theory activities and coding workshops. I am continuing to practice so that I can improve my ability to write, understand, debug, and explain small programs.

4. Tools and Learning Resources

The following table summarizes the learning resources I used or explored my skill-based training to becoming a good medical professional who is smart in clinical, theory, and technologically to run with latest technological devices, and protect patients' data securely.

Resource	Purpose	Evidence
INSA Cyber Talent Summer Camp Challenge	Structured skill-based training.	https://ctc.insa.gov.et/
TryHackMe	Guided cybersecurity rooms and practical labs.	https://tryhackme.com/p/solomongetu2?
PicoCTF	Independent practice through cybersecurity challenges.	https://learn.cylabacademy.org/profile

freeCodeCamp — Python	Introductory programming concepts and coding workshops.	https://www.freecodecamp.org/learn/python-v9/
freeCodeCamp — English A2 material	Vocabulary and comprehension development for learning.	https://www.freecodecamp.org/learn/a2-english-for-developers/
Nmap documentation	Reference for scan options and command behavior.	https://nmap.org/book/man.html
Burp Suite documentation	Reference for web application testing features.	https://portswigger.net/burp/documentation
Kali Linux documentation	Reference for Linux distribution and tools.	https://www.kali.org/docs/
OWASP	Web application security learning and references.	https://owasp.org /

- In addition, There were other materials provided by mentor, such as slides, demonstrations, or course notes.

5. Practical Work and Technical Documentation

I prepared PDF write-ups to record selected technical topics and practical exercises. These documents are part of the supporting evidence for this portfolio and will be linked from the main GitHub repository.

Write-up / area	What to document	
OWASP-related write-up	Topic, objective, key concepts, methodology, observations, lessons, and references.	**https://github.com/solomongetu2-max/Cybersecurity-skills-portfolio/blob/main/**
Airplane lab	Lab objective, authorized environment, approach, relevant findings, and lessons learned.	
Blue lab	Lab objective, methods, observations, and lessons learned, consistent with platform sharing rules.	
Networking	Concepts practiced, commands used, output interpretation, and troubleshooting.	
Python code	Purpose of each program, source code, sample output, and explanation.	

- For each substantial lab report, I aim to explain why each important step was performed, what the output indicated, what difficulties occurred, and what I learned.

6. Independent Learning and Personal Development

6.1 Independent Cybersecurity Practice

In addition to the formal training, I practiced independently using TryHackMe and PicoCTF. This gave me additional opportunities to revisit concepts, attempt challenges, investigate errors, and build persistence when solving unfamiliar problems.

6.2 Python Learning

I used freeCodeCamp to study Python fundamentals, including introductory concepts and selected practical coding workshops. I view programming as a useful supporting skill for cybersecurity, including for automation, data processing, and understanding scripts. I will continue practicing rather than treating course progress alone as proof of mastery.

6.3 English-Language Development

I also studied introductory English material at A2 level to improve vocabulary and comprehension. This is relevant because cybersecurity learning often involves reading technical documentation, instructions, reports, and discussions. I plan to continue improving my English alongside technical skills.

7. Evidence of Learning

The portfolio will include selected evidence that supports the activities described above. Evidence may include screenshots of course progress, relevant lab terminal output, code and program output, and excerpts from technical write-ups. Each image will have a short caption explaining what it demonstrates.

8. Challenges and Lessons Learned

An important lesson from this learning period is that understanding a concept and applying it independently are different stages of development. I have gained exposure to foundational topics and practical tools, but regular repetition is still necessary to improve accuracy, confidence, and problem-solving speed.

Writing technical reports also showed me the importance of documenting each step clearly, explaining the reason behind commands, interpreting results, and consulting reliable references. When an activity does not work as expected, troubleshooting systematically is an important part of learning.

I intend to use future practice to strengthen weak areas, revisit difficult concepts, and become more independent when approaching unfamiliar lab tasks.

9. Future Learning Goals

- Strengthen Linux command-line skills, file permissions, processes, and networking commands.
- Improve understanding of networking fundamentals, ports, protocols, services, and traffic analysis.

- Practice interpreting Nmap results and using security tools responsibly in authorized environments.
- Develop a clearer understanding of web application security concepts and safe testing methodology.
- Continue solving TryHackMe and PicoCTF challenges, recording lessons learned and distinguishing completed work from work in progress.
- Improve Python programming through regular coding practice and small projects.
- Improve technical English vocabulary and reading comprehension.
- Produce well-structured write-ups with objectives, methodology, evidence, explanations, and references.
- Hopefully, I will be a great physician, and Cybersecurity expert specifically in cloud security.

10. Conclusion

The Cyber Talent Summer Camp Challenge provided an important foundation for my continued development in cybersecurity. Through training, laboratory practice, technical write-ups, and independent learning, I explored Linux, networking, security tools, Python programming, and supporting English-language skills.

I understand that cybersecurity competence develops through continued practice, careful analysis, ethical conduct, and ongoing learning. I will continue strengthening my practical skills and improving the quality of my documentation as I work toward greater independence and technical proficiency.

End of report